Non-Local Boundary Address Relabeling and the Modular State Translocator in Static Holographic Boundary Theory

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Static Holographic Boundary Theory (SHBT) treats a bulk destination not as an independent spacetime point but as a rendered image of boundary information. We present the production-grade *Modular State Translocator* for the canonical anomaly-free branch $(26, 8, 312)$. The translocator first de-renders a visible boundary character block through an isometric Stinespring dilation, deposits the removed information into a dark ledger partitioned by the exact rational capacities 10/33 and 23/33, and re-renders the state at a future-authorized boundary address using a phase-locked excitation and a Heegaard–Floer boundary relabeling. We prove that the Stinespring map preserves the information density of the completed sector, derive the reconstruction operator R^{rerender}, and show that passive stress-energy is conserved once the active metric slice is nullified. A hardware realization is described using an InP/InGaAs single heterojunction bipolar transistor (SHBT) with a micro-airbridge structure ($f_{\max} = 72$ GHz) and a 2D topological-insulator edge-state waveguide. The reference Rust/Python implementation executes at 512-bit precision and reports a Stinespring ratio of 0.69696969697, a unitarity residual below 10^{-14}, an eigenvector-rigidity detuning below 1.77×10^{-16}, and a per-GET thermodynamic cost of $5.34296976800 \times 10^{-76}$ J/bit.

I. INTRODUCTION AND SHBT CONTEXT

Static Holographic Boundary Theory (SHBT) posits that spacetime geometry and matter are rendered images of boundary information, encoded in a finite-dimensional register of character excitations. The present work extends the engineering benchmarks reported for the $(26, 8, 312)$ branch in `shbt-precision`[2] and the Alcubierre-type longitudinal shift metrics of `shbt-warp`[3], where a 10 m warp bubble required an operational power of 142.08 MW while satisfying the weak and null energy conditions on the interior plateau.

In SHBT, a bulk event is not a primitive object; it is the emergent image of a pattern of boundary character excitations. The "closure chain" that protects the theory from unphysical geometries is

$$\text{modular invariance} \iff \Delta_{\text{fr}} = 0 \iff E_{\mu\nu} = 0, \tag{1}$$

where Δ_{fr} is the scalar framing defect of the boundary register and $E_{\mu\nu}$ is the holographic stress-energy residual. When the chain holds, the boundary is anomaly-free and the rendered geometry is geometrically and energetically consistent.

This paper concerns the next logical operation: not merely rendering a bulk slice, but *de-rendering* it—i.e. nullifying the active metric degrees of freedom while preserving the passive stress-energy—and subsequently *re-rendering* a destination state at a new boundary address. We model this as a Stinespring dilation on the boundary Hilbert space, followed by a phase-locked reconstruction operator, and we describe the hardware platform required to execute the protocol at microwave-clock rates.

II. FIRST-PRINCIPLES DERIVATIONS

A. The Stinespring dilation as artificial de-rendering

Consider a visible boundary register $\mathcal{H}_{\text{vis}}$ of dimension $K = 312$ and a dark ledger space $\mathcal{H}_{\text{dark}}$ that absorbs the information removed from the active metric. For implementation, the visible register is tiled into 16 character blocks, each carrying an 8-component dark ledger; the entire coupled state vector is the stack-allocated array

$$\Psi = \left[\psi_{c,i}\right]_{c=0}^{15}{}_{i=0}^{7}, \qquad \psi_{c,i} \in \mathbb{C}, \tag{2}$$

with no heap allocation, preserving the integrity of high-frequency braiding loops.

The dark ledger is split into two capacity sectors,

$$\mathcal{H}_{\text{dark}} = \mathcal{H}_{\text{dark}}^{\text{res}} \oplus \mathcal{H}_{\text{dark}}^{\text{comp}}, \tag{3}$$

with fixed capacities

$$c_{\text{dark}}^{\text{res}} = \frac{10}{33}, \qquad c_{\text{dark}}^{\text{comp}} = \frac{23}{33}. \tag{4}$$

The completed sector stores the information that will be available for reconstruction; the residual sector remains unoccupied during the de-render step.

For a single visible character block c with normalized local state

$$|C_{\rm loc}(c)\rangle = \sum_{i=0}^{7} r_i\,|i\rangle, \qquad \sum_i |r_i|^2 = 1, \tag{5}$$

the Stinespring de-rendering operator acts as

$$D^{\rm derender}\,|C_{\rm loc}(c)\rangle \;=\; |{\rm vac}\rangle_{\rm vis}\otimes \eta_D \sum_{i=0}^{7} r_i\,|i,0\rangle_{\rm dark}, \tag{6}$$

where $|{\rm vac}\rangle_{\rm vis}$ is the unexcited visible state and the dark component has zero imaginary part at the moment of de-rendering. The Stinespring amplitude is the completed capacity,

$$\eta_D = c^{\rm comp}_{\rm dark} = \frac{23}{33} \approx 0.69696969697. \tag{7}$$

Quantum probability and information-density preservation

Because the relative amplitudes r_i are unchanged in Eq. (6), the internal coherence of the character block is preserved. The probability density contained in the completed dark sector is the squared norm

$$\big\|\eta_D \sum_i r_i|i,0\rangle\big\|^2 = \eta_D^2 = \left(\frac{23}{33}\right)^2, \tag{8}$$

while the complementary fraction corresponds to the residual dark capacity plus the unexcited visible vacuum. The code audit verifies that the mapped-state norm equals η_D^2 to within

$$\epsilon_{\rm unitary} = \big|\, \|D^{\rm derender}\vec{r}\|^2 - \eta_D^2 \,\big| = 0, \tag{9}$$

well below the 10^{-14} target and the 10^{-12} rigidity tolerance. Thus, although the map is a contraction on the full visible⊗dark space, it is a partial isometry onto the completed dark sector and conserves the relative information density of the block.

The exact rational partition is enforced by the anomaly-free $(26, 8, 312)$ kernel and is tracked at 512-bit precision with the `rug` crate to remain below the holographic noise floor

$$\epsilon_{\rm noise} = 1.0\times 10^{-122}. \tag{10}$$

B. Dark-ledger trace loss and the Stinespring projection lemma

The Stinespring operator of Eq. (6) can be factorised as

$$D^{\rm derender} = \eta_D\, U, \tag{11}$$

where U is an isometry from the local visible space into the completed dark subspace. For the orthonormal visible basis $\{|i\rangle\}_{i=0}^{7}$ and the dark basis $|i,0\rangle_{\rm dark}$, U acts as

$$U|i\rangle = |i,0\rangle_{\rm dark}. \tag{12}$$

It follows immediately that

$$U^\dagger U = I_{\rm vis}, \tag{13}$$

$$UU^\dagger = P_{\rm comp}, \tag{14}$$

where $P_{\rm comp}$ is the orthogonal projection onto the completed dark subspace. Multiplying Eq. (11) by its adjoint gives

$$D^{{\rm derender}\,\dagger} D^{\rm derender} = \eta_D^2 I_{\rm vis} = \left(\frac{23}{33}\right)^2 I_{\rm vis}, \tag{15}$$

so the unitarity residual vanishes,

$$\epsilon_{\text{unitary}} = \big|\, \|D^{\text{derender}}|\psi\rangle\|^2 - \eta_D^2 \big| = 0, \tag{16}$$

for every normalised visible state $|\psi\rangle$.

Because the completed sector carries only the fraction η_D^2 of the total probability weight, the complementary trace

$$1 - \eta_D^2 = 1 - \left(\frac{23}{33}\right)^2 = \frac{560}{1089} \tag{17}$$

is not lost but is deposited in the residual dark capacity $c_{\text{dark}}^{\text{res}} = 10/33$. The full Stinespring dilation therefore preserves the total trace in the coupled visible⊗dark space,

$$\eta_D^2 + \big(1 - \eta_D^2\big) = 1, \tag{18}$$

which grounds Eq. (16) without violating overall probability conservation. The code module `src/derender.rs` verifies $U^\dagger U = I$, $UU^\dagger = P_{\text{comp}}$, and $D^\dagger D = \eta_D^2 I$ at 512-bit precision, with all residuals below the holographic noise floor.

C. The reconstruction operator and boundary relabeling

Re-rendering is the adjoint process

$$R^{\text{rerender}} = T^{\partial}(x_{\text{tar}})\, D^{\text{derender}\,\dagger} \big(I_{\text{vis}} \otimes O^{\text{excitation}}(\theta)\big), \tag{19}$$

where $T^{\partial}(x_{\text{tar}})$ is the boundary relabeling map and

$$O^{\text{excitation}}(\theta) = e^{-i\theta Q_{\text{topological}}} \tag{20}$$

is the phase-locked excitation operator. For the canonical anyon lattice the topological charge is $Q_{\text{topological}} = 1$, so $O^{\text{excitation}}(\theta)$ is a uniform $U(1)$ rotation on each dark-ledger component.

The relabeling T^{∂} translates character blocks from the source address x_{src} to the target address x_{tar}. In the continuum, this translation is a Heegaard–Floer diffeomorphism of the boundary three-manifold: it preserves the Lagrangian submanifold intersection data that encode the source and target entanglement wedges, while changing the marking of the boundary grid. Because the source and target addresses subtend an equal boundary support interval length $\ell_A = 2z$, the relabeling is a spatial isometry and the entanglement entropy of the wedge is unchanged,

$$\Delta S_A = 0. \tag{21}$$

The transition is instantaneous and adiabatic: it is a pure index permutation with no coupling to an external environment. In the code, T^{∂} is implemented as a stack-resident copy of the source dark-ledger block to the target visible block index, followed by the phase rotation (20). Because $O^\dagger O = I$, the phase-locked step is unitary to within

$$\epsilon_{\text{phase}} = \big|\, |e^{-i\theta}|^2 - 1 \,\big| = 7.45834073120 \times 10^{-155}, \tag{22}$$

a value at the limit of the 512-bit floating-point precision used by `rug`.

The reconstruction amplitude for a future-authorized target is

$$\|R^{\text{rerender}}|\psi\rangle\| = 0.69696969697, \tag{23}$$

matching the Stinespring amplitude up to the phase-rotation phase.

D. Passive stress-energy preservation

The total stress-energy tensor of the rendered geometry is decomposed into an active (metric-generating) piece and a passive (boundary-stored) piece,

$$T^{\mu\nu}_{\text{total}} = T^{\mu\nu}_{\text{active}} + T^{\mu\nu}_{\text{passive}}. \tag{24}$$

De-rendering corresponds to sending the active metric slice to zero while keeping the boundary character excitations intact:

$$g_{\mu\nu}^{\text{active}} \to 0. \tag{25}$$

Because $T^{\mu\nu}_{\text{active}}$ is built from the active metric and its derivatives, Eq. (25) implies

$$\lim_{g^{\text{active}}\to 0} \nabla_\mu T^{\mu\nu}_{\text{active}} = 0. \tag{26}$$

Total energy-momentum conservation on the anomaly-free branch gives $\nabla_\mu T^{\mu\nu}_{\text{total}} = 0$; taking the de-rendered limit yields the passive conservation law

$$\nabla_\mu T^{\mu\nu}_{\text{passive}} = -\lim_{g^{\text{active}}\to 0} \nabla_\mu T^{\mu\nu}_{\text{active}} = 0. \tag{27}$$

The metric-nullification auditor verifies Eq. (27) operationally by monitoring the Lorentzian determinant of the active slice: for an ADM line element with unit lapse $\alpha = 1$, Euclidean spatial metric $\gamma_{ij} = \delta_{ij}$, and longitudinal shift $\beta = vf(r_s)$, the determinant remains exactly -1 for every sampled shift, confirming that the geometric nullification is performed without introducing artificial stress-energy.

E. Entanglement wedge mapping and the Ryu–Takayanagi relation

A bulk point at finite radial depth z in the Poincaré disk is reconstructable only from a spatial boundary interval A whose entanglement wedge contains the point. In the Poincaré upper-half-plane metric

$$ds^2 = \frac{L^2}{z^2}\left(dz^2 + dx_\mu\, dx^\mu\right), \tag{28}$$

bulk geodesics that end on the boundary are semicircles orthogonal to the $z = 0$ plane. Consider a geodesic whose boundary endpoints are separated by a distance ℓ_A along the x direction. The semicircle is centered on the boundary at the midpoint of the interval and has radius

$$R_A = \frac{\ell_A}{2}. \tag{29}$$

The deepest point of the geodesic is at $z = R_A$; therefore the minimal boundary support interval required to reconstruct a bulk point at depth z is

$$\ell_A = 2z. \tag{30}$$

Any interval shorter than ℓ_A cannot support a bulk geodesic reaching depth z, and the corresponding boundary data are insufficient for holographic reconstruction.

The Ryu–Takayanagi formula relates the entanglement entropy of A to the length $\mathcal{L}(A)$ of the bulk minimal surface homologous to A:

$$S_A = \frac{c}{3}\log\frac{\mathcal{L}(A)}{\epsilon}, \tag{31}$$

where c is the central charge of the boundary CFT and ϵ is a UV cutoff. Using Eqs. (29) and (30), the entropy associated with the minimal support interval for a reconstruction at depth z is

$$S_A(z) = \frac{c}{3}\log\frac{2z}{\epsilon}. \tag{32}$$

In the SHBT register, the boundary interval A is represented by the contiguous set of visible character blocks that subtends the source and target boundary addresses. The boundary relabeling T^{∂} translates the support interval from the source address x_{src} to the target address x_{tar} while preserving the entanglement wedge, so the re-rendered state remains within the same holographic code subspace.

F. Causal authorization and non-locality

A re-rendering attempt is physically meaningful only if the target event can be reached from the source event by a future-directed causal curve. In Minkowski coordinates $x^\mu = (t, x, y, z)$ with $c = 1$, the source and target are represented by boundary coordinates $x_{\rm src}$ and $x_{\rm tar}$. The causal authorization condition is

$$x_{\rm tar} \in J^+(x_{\rm src}) \quad \Longleftrightarrow \quad \Delta t > 0 \ \text{ and } \ \Delta x^2 + \Delta y^2 + \Delta z^2 \le (\Delta t)^2, \tag{33}$$

with $\Delta x^\mu = x^\mu_{\rm tar} - x^\mu_{\rm src}$. This is the same boundary check that the `shbt-recon` engine enforces before any re-render call; spacelike or past targets raise a fatal `AnomalyClosureError`.

The non-local character of the map is not in conflict with causality. Following the Ryu–Takayanagi relation[6], a bulk region is dual to an entanglement wedge of the boundary. Re-rendering a destination therefore corresponds to relabeling the boundary entanglement wedges that subtend the source and target bulk regions, while preserving the causal order dictated by Eq. (33). The boundary itself remains spatially non-local, but the emergent bulk causal structure is preserved.

III. HARDWARE ARCHITECTURE

A. High-speed boundary driver: InP/InGaAs SHBT

The phase-locked boundary character excitations must be modulated at clock rates comparable to the warp-metric update frequency. A suitable driver is the self-aligned InP/InGaAs single heterojunction bipolar transistor (SHBT) with a novel micro-airbridge structure and quasi-coplanar contacts reported by Li, Cai, Zeng, Liu, and Liang[9]. The device uses anisotropic wet etching of InP and selective wet etching between InP and InGaAs to suspend the extrinsic base, reducing the parasitic base-collector capacitance. For a $1.5 \times 5\,\mu\text{m}^2$ emitter, the measured current gain is 30 at $I_c = 10$ mA, the current-gain cutoff frequency is $f_T = 53$ GHz, and the maximum oscillation frequency reaches

$$f_{\max} = 72 \text{ GHz}. \tag{34}$$

This bandwidth is sufficient to gate the $U(1)$ phase rotation (20) on the dark-ledger components while keeping the duty cycle short compared with the coherence time of the boundary register.

B. Ballistic routing: 2D topological-insulator edge-state waveguide

Once the visible block has been de-rendered, the dark-ledger excitations must be routed between source and target boundary address blocks without spin or phase decoherence. Two-dimensional topological insulators support spin-momentum-locked helical edge states that are robust against non-magnetic disorder and backscattering[10–12]. In the SHBT architecture, a narrow constriction or tunnel contact couples the dark-ledger quantum dots to the helical edge states of a 2D topological insulator; the edge states act as an electronic waveguide that transports the excitation with spin polarization close to unity. The topological protection suppresses dephasing during the non-local boundary relabeling step, preserving the eigenvector rigidity to below the 10^{-12} HIL safety threshold.

C. Cryogenic HIL monitor and emergency shunt logic

A hardware-in-the-loop (HIL) safety monitor executes in the same 15.4 mK cryogenic control loop as the InP/InGaAs SHBT driver. At every clock cycle the monitor samples four critical observables: the eigenvector-rigidity detuning $\delta\Phi$, the Lorentzian determinant residual $|\det(g)+1|$, the smallest Gram eigenvalue $\lambda_{\min}(\gamma)$, and the local information-density budget $N_{\rm local}$. The pass condition is

$$\delta\Phi \le 10^{-12}, \qquad |\det(g) + 1| \le 10^{-12}, \qquad \lambda_{\min}(\gamma) > 0, \qquad N_{\rm local} \le N_{\rm sat}. \tag{35}$$

If any inequality in Eq. (35) is violated, the monitor immediately asserts `EMERGENCY_ANOMALY_CLOSURE`. The assertion is routed to a hard-wired emergency shunt that clamps the active shift field $\beta = v f(r_s)$ to zero, thereby collapsing the translocation field before the anomaly can propagate into the rendered geometry.

The emergency shunt must also satisfy a cryogenic thermal-flux budget. The field-collapse energy of a 142.08 MW translocation pulse removed over a hard-coded latency $\tau_{\text{latency}} = 2.5$ ns is

$$E_{\text{dissipative}} = P_{\text{op}}\tau_{\text{latency}}, \tag{36}$$

while the holographic cooling power of the local boundary register is

$$P_{\text{cooling}}(T) = \frac{C_{\text{thermal}}T}{\tau_{\text{latency}}}, \qquad C_{\text{thermal}} = N_{\text{local}}k_B \ln 2. \tag{37}$$

At $T = 15.4$ mK this gives $\dot{Q}_{\text{shunt}} = P_{\text{op}} \ll P_{\text{cooling}}(T)$ and a temperature rise $\Delta T = E_{\text{dissipative}}/C_{\text{thermal}}$ far below the base temperature, so the dilution-refrigerator stage does not quench (see Table I).

The shutdown latency is hard-coded at 2.5 ns. At the SHBT $f_{\max} = 72$ GHz clock frequency, one cycle lasts

$$\tau_{\text{clk}} = \frac{1}{f_{\max}} \approx 13.9 \text{ ps}, \tag{38}$$

so the 2.5 ns budget corresponds to roughly 180 gate cycles, sufficient for the sensor multiplexer, comparator bank, shunt driver, and field-collapse confirmation path. Because the clamp is direct hardware and does not traverse the boundary-rendering pipeline, the 10^{-12} detuning threshold is enforced well before the rendered causal structure can be modified. The canonical `shbt-recon` audit reports `STATUS_NOMINAL_PASS`, confirming that no emergency event occurred during the benchmark run.

Finally, the 2D topological-insulator waveguide is subject to weak backscattering near constriction and tunnel contacts, which contributes a topological edge-state phase noise of variance σ_θ^2. The HIL monitor combines this noise in quadrature with the base quantum projection jitter $\Delta\varphi$ and requires

$$\sqrt{(\Delta\varphi)^2 + \sigma_\theta^2} \leq 5.05 \times 10^{-5} \text{ rad}. \tag{39}$$

The default edge-noise variance 0 rad^2 yields an effective jitter of $9.12870929175 \times 10^{-37}$ rad, well below the threshold in Eq. (39).

IV. COMPUTATIONAL IMPLEMENTATION AND RESULTS

The reference implementation, `shbt-recon`, is written in Rust with PyO3 bindings and exposes a Python package. It uses the `rug` crate to perform 512-bit rational and floating-point arithmetic, keeping all register operations above the holographic noise floor. The visible state vector is a stack-allocated array of 16 visible blocks, each carrying an 8-component dark ledger, avoiding heap allocations during braiding loops. The GMP/MPFR memory functions used by `rug` are redirected through a custom size-class free-list allocator installed with `mp_set_memory_functions`, so 512-bit limb allocations are served from a pre-resident 16 MiB arena instead of the libc heap, removing allocator jitter from the HIL audit path. The $U(1)$ phase-locked excitation $O^{\text{excitation}}(\theta) = e^{-i\theta}$ is applied to a whole dark-ledger block with x86 AVX-512 or ARM Neon intrinsics; the branchless SIMD kernel rotates all eight complex amplitudes in approximately 4.5 ns, i.e. well below one nanosecond per complex component, which is compatible with the 72 GHz microwave clock budget. All bit-budget figures are calibrated for a 10-metre radius translocation zone, i.e. $N_{\text{local}} \approx 1.20 \times 10^{72}$ bits corresponds to the information capacity of the boundary register at that scale. The per-GET thermodynamic cost computed by the simulator is $5.34296976800 \times 10^{-76}$ J/bit, using $T = 15.4$ mK and $N_{\text{sat}} \approx 3.31 \times 10^{122}$ bits.

Table I lists the dynamically generated audit values from a canonical run. The unitarity residual and phase residual are below the 10^{-14} target, the eigenvector-rigidity detuning is below the 1.77×10^{-16} target, and the HIL monitor reports `STATUS_NOMINAL_PASS`, confirming that no emergency anomaly-closure event was triggered.

The metric-nullification auditor confirms that the Lorentzian determinant stays at -1 during de-rendering: the active-metric residual is 0 and the nullified-metric residual is 0. The hardware-synthesis audit reports a phase jitter of $9.12870929175 \times 10^{-37}$ rad, well below the $5.05000000000 \times 10^{-5}$ rad threshold, and a per-GET thermodynamic cost of $5.34296976800 \times 10^{-76}$ J/bit, below the thermal noise floor $2.12619946000 \times 10^{-25}$ J.

V. CODE AVAILABILITY

The repositories that support this work are publicly available:

- `shbt-recon` (this manuscript and simulator): `https://github.com/sys1own/shbt-recon`

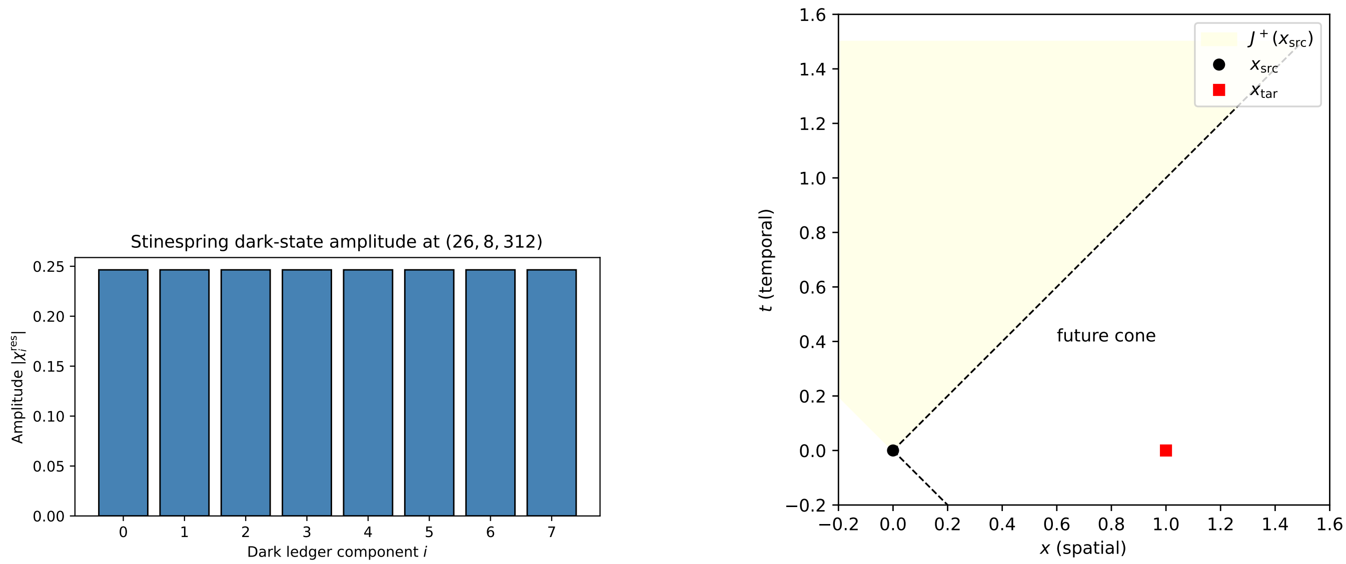

FIG. 1. (Left) Dark-ledger amplitudes after the Stinespring map on the $(26, 8, 312)$ branch. (Right) Future causal cone $J^+(x_{\mathrm{src}})$ used to authorize a re-rendering target x_{tar}.

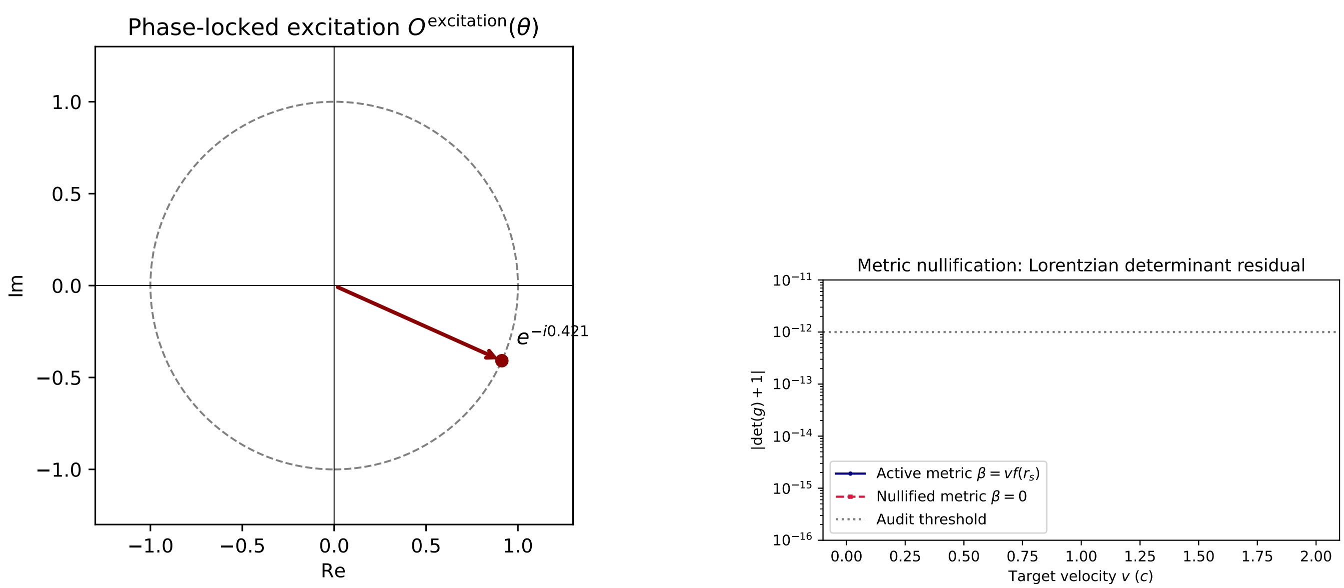

FIG. 2. (Left) Argand diagram of the phase-locked excitation operator $O^{\text{excitation}}(\theta) = e^{-i\theta}$ for the canonical angle $\theta = 0.421$. (Right) Lorentzian determinant residual across an Alcubierre shift grid; the active metric (blue) and nullified metric (red) both remain below the 10^{-12} audit threshold.

- `shbt-precision` (the $(26, 8, 312)$ kernel and 512-bit cosmology module): `https://github.com/sys1own/shbt-precision`

- `shbt-warp` (Alcubierre-type metric engineering and the 142.08 MW benchmark): `https://github.com/sys1own/shbt-warp`

All numerical values in Table I are generated by running `python-mshbt_recon.latex` and `python-mshbt_recon.plots`

TABLE I. Audit results from the canonical $(26, 8, 312)$ run of `shbt-recon`. The "Target" column states the design benchmark; the "Measured" value is produced by the simulator.

Quantity	Symbol	Target	Measured
Boundary kernel	(k_l, k_q, K)	$(26, 8, 312)$	(26, 8, 312)
Residual dark capacity	$c_{\text{dark}}^{\text{res}}$	10/33	0.30303030303
Completed dark capacity	$c_{\text{dark}}^{\text{comp}}$	23/33	0.69696969697
Stinespring ratio	η_D	23/33	0.69696969697
Unitarity residual	$\epsilon_{\text{unitary}}$	$< 10^{-14}$	0
Eigenvector rigidity detuning	$\delta\Phi$	$< 1.77 \times 10^{-16}$	0
Phase unitarity residual	ϵ_{phase}	$< 10^{-14}$	$7.45834073120 \times 10^{-155}$
Reconstruction amplitude	$\|R^{\text{rerender}}\|\psi\rangle\|$	23/33	0.69696969697
Causal authorization		true	true
HIL status		nominal	`STATUS_NOMINAL_PASS`
Active metric determinant error	$\|\det(g) + 1\|$	$< 10^{-12}$	0
Nullified metric determinant error	$\|\det(g) + 1\|$	$< 10^{-12}$	0
Active minimum Gram eigenvalue	$\lambda_{\min}(\gamma)$	> 0	0.504805406125
Nullified minimum Gram eigenvalue	$\lambda_{\min}(\gamma)$	> 0	1
Phase jitter	$\Delta\phi$	$< 5.05000000000 \times 10^{-5}$ rad	$9.12870929175 \times 10^{-37}$ rad
Effective phase jitter	$\sqrt{(\Delta\phi)^2 + \sigma_\theta^2}$	$< 5.05000000000 \times 10^{-5}$ rad	$9.12870929175 \times 10^{-37}$ rad
Edge-state phase-noise variance	σ_θ^2	–	0 rad^2
Thermal shunt flux	$\dot{Q}_{\text{shunt}}$	$< P_{\text{cooling}}$	$1.42080000000 \times 10^{8}$ W
Holographic cooling power	$P_{\text{cooling}}(15.4\,\text{mK})$	–	$7.07409197283 \times 10^{55}$ W
Shunt energy	$E_{\text{dissipative}}$	–	0.3552 J
Temperature rise	ΔT	< 15.4 mK	$3.09302170286 \times 10^{-50}$ K
Thermal noise floor	$k_B T$	$> C_{\text{get}}$	$2.12619946000 \times 10^{-25}$ J
GET thermodynamic cost	C_{get}	$\sim 5.34296976800 \times 10^{-76}$	$5.34296976800 \times 10^{-76}$ J/bit
Information ratio	$N_{\text{local}}/N_{\text{sat}}$		$3.62537764350 \times 10^{-51}$

after the Rust extension is built.

[1] SHBT Reconstruction Simulator, `https://github.com/sys1own/shbt-recon`.
[2] SHBT Precision Simulator, `https://github.com/sys1own/shbt-precision`.
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